

Rui Casaca

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Santa Maria da Feira, Portugal

About me

Junior Software Engineer completing a B.Sc. in Computer Engineering at ISEC, with professional experience in React, TypeScript and Python gained through an internship at Critical Software.

Interested in building reliable software systems, full-stack applications, real-time platforms and offline-first mobile solutions that solve real-world problems.

Technical Skills

Languages: TypeScript, Python, Java, C/C++, C#

Frontend: React, HTML/CSS

Mobile: Flutter, Kotlin

Backend: Node.js

Familiar With: C#, ASP.NET Core (.NET 8), Entity Framework Core

Databases: SQLite, MongoDB, Firebase

Tools: Git, GitHub, BitBucket, Docker, Postman, Confluence

Concepts: REST APIs, Micro-Services, Authentication, Agile, Software Validation, Requirements

Professional Experience

Software Engineering Intern

Critical Software — Feb 2026 - Jul 2026

- Developed frontend components in React and TypeScript for LightHouse, a Ground Segment platform for space systems.
- Implemented features for real-time telemetry visualization and processing through integration with REST APIs and WebSockets.
- Contributed to frontend and backend development, including Python-based services, while collaborating on requirements analysis and feature design.
- Participated in system validation activities through functional testing, API testing and defect analysis.

Freelance GIS Specialist & 3D Modeling

Fiverr, upwork — Nov 2022 - Present

- Developed GIS solutions, data-driven applications and 3D modelling projects, translating technical and operational requirements into functional solutions.
- Coordinated small teams and ensured project delivery quality.

Military & GIS Analyst and Operations Technician

GNR – UEPS (National Republican Guard – Emergency Protection & Relief Unit) — Oct 2011 - Mar 2019, Mar 2019 - Feb 2026

- Supported missing persons search and disaster response operations through the use of drones and Geographic Information Systems (GIS).
- Worked in mission-critical environments requiring rapid decision-making, teamwork and operational accuracy.

Projects

HomeVault - Offline-First Inventory Management App

Flutter, Riverpod, Drift (SQLite), Google Drive API

- Developed an Android home inventory management application with an offline-first architecture, enabling users to catalog assets, documents and photos securely.
- Implemented local data management with Drift (SQLite), encrypted backups and Google Drive synchronization, biometric authentication, QR/barcode scanning and Excel export.

Missing Persons Search Application

React, TypeScript, Node.js, LLM APIs

- Developed a full-stack platform supporting missing persons search operations, enabling centralized information management and analysis. Integrated LLM APIs for contextual data analysis, implemented real-time dashboards and automated PDF report generation with email distribution.

Teacher and Student Evaluation Platform (Academic)

Java, Client-Server Architecture, RMI, SQLite

- Developed a client-server application for teacher and student evaluation processes, including authentication, role-based access control, questionnaire management, result processing and SQLite-based data storage.

Education

Bachelor's in Computer Engineering – Software Development Track

Instituto Superior de Eng. de Coimbra (ISEC) — 2023 - 2026